인공지능수학기초 (01)

Final Exam
June 17, 2025

1. (4+4 pts) 두 개의 random variable X와 Y의 joint pdf가 아래와 같을 때, 다음을 구하시오.

$$f_{X,Y}(x,y) = kx^2y^2e^{-y},\ 0 < x < 1,\ 0 < y < \infty.$$

   (a) k를 구하시오.
   (b) X와 Y의 marginal pdf를 각각 구하시오.

2. (4+4 pts) Random variable X가 다음과 같이 gamma 분포를 따를 때, 다음을 구하시오.

$$f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}},\ (\alpha > 0,\ \beta > 0,\ and\ \ x > 0)$$

   (a) $E(e^{tX})$ 를 구하시오.
   (b) (a)를 이용해서, $E(X)$를 구하시오.

3. (4+6+4 pts) Random variable X 와 Y에 관한 trinomial을 보자.

$$f(x,y) = \frac{n!}{x!\,y!\,(n-x-y)!} p^x q^y (1-p-q)^{n-x-y},$$

$$x, y = 0, 1, \ldots, n,\ \ x + y \le n,\ \ p, q > 0\ \ \ p + q < 1.$$

   (a) X에 관한 marginal pmf를 구하시오.
   (b) $(X+Y)$의 분포를 구하시오.
   (c) $(X+Y)|X$의 분포를 구하시오.

4. (4 pts) 다음의 분포가 정상적인 PDF임을 보이시오.

$$f_X(x) = \frac{\lambda^k x^{k-1}}{(k-1)!} e^{-\lambda x},\ \ x > 0\ \ \&\ \ k = 1, 2, 3, \cdots$$

5. (6 pts) 서로 독립인 확률 변수 $X \sim$ unif $(0,1)$이고, $Y\sim$ unif$(0,1/10)$이다. $Z = X — Y$일 때, X와 Z의 correlation coefficient를 구하시오. (unif(a,b)는 a에서 b 까지의 uniform distribution)

1. $f_{X,Y}(x,y) = (ax^2)(by^2e^{-y}) \leftarrow$ Factoritable

$ab = k. \quad \rightarrow$ Indep.

$f_X(x) = ax^2 \qquad \int_0^1 ax^2\,dx = a/3 = 1 \leftarrow a = 3$

$f_Y(y) = by^2e^{-y} \qquad \int_0^\infty by^2e^{-y}\,dy = 1$

$b\Gamma(3) = 2b = 1 \leftarrow b = 1/2$

a) $k = \frac{3}{2}$

b) $f_X(x) = 3x^2$

$f_Y(y) = 1/2\, y^2 e^{-y}$

2. a) $E(e^{tX}) = \int_0^\infty e^{tx} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}}\,dx$

$$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty x^{\alpha-1} e^{-\frac{x}{\beta}(1-\beta t)}\,dx$$

$\frac{(1-\beta t)}{\beta}x = y \qquad x = \frac{\beta}{1-\beta t}y \qquad dx = \frac{\beta}{1-\beta t}dy$

$\llcorner \ (1-\beta t) > 0$

$\rightarrow t < 1/\beta.$

$$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty \left(\frac{\beta}{1-\beta t}y\right)^{\alpha-1} e^{-y} \cdot \frac{\beta}{1-\beta t}\,dy$$

$$= \frac{1}{\Gamma(\alpha)(1-\beta t)^\alpha} \underbrace{\int_0^\infty y^{\alpha-1}e^{-y}\,dy}_{=\Gamma(\alpha)}$$

$= \left(\frac{1}{1-\beta t}\right)^\alpha \quad (t < 1/\beta) \leftarrow$ 없으면 1점 감점

b) $E(X) = \frac{d}{dt} E(e^{tX}) \Big|_{t=0}$

$$= \frac{(-\alpha)\times(-\beta)}{(1-\beta t)^{-\alpha+1}} \Big|_{t=0} = \alpha\beta.$$

3. a) $f_X(x) = \sum_{y=0}^{n-x} f_{X,Y}(x,y)$

$$= \frac{n!}{x!} p^x \sum_{y=0}^{n-x} \frac{1}{y!(n-x-y)!} q^y (1-p-q)^{n-x-y} \cdot \frac{(n-x)!}{(n-x)!}$$

$$= \frac{n!}{x!(n-x)!} p^x \underbrace{\sum_{y=0}^{n-x} \binom{n-x}{y} q^y (1-p-q)^{n-x-y}}_{=(1-p)^{n-x}}$$

$$= \binom{n}{x} p^x (1-p)^{n-x} \to \text{bin}(n,p)$$

b) Let $U = X+Y$, $V = X$.

Then, $X = V$, $Y = U - V$.

$$f_{U,V}(u,v) = \frac{n!}{v!(u-v)!(n-u)!} p^v q^{u-v} (1-p-q)^{n-u}$$

$$f_U(u) = \sum_{v=0}^{u} f_{U,V}(u,v) \quad (\because y = u-v \geq 0)$$

$$= \frac{n!}{(n-u)!} (1-p-q)^{n-u} \sum_{v=0}^{u} \frac{1}{v!(u-v)!} p^v q^{u-v} \cdot \frac{u!}{u!}$$

$$= \frac{n!}{u!(n-u)!} (1-p-q)^{n-u} \underbrace{\sum_{v=0}^{u} \binom{u}{v} p^v q^{u-v}}_{=(p+q)^u}$$

$$= \binom{n}{u} (p+q)^u (1-p-q)^{n-u} \to \text{bin}(n, p+q)$$

c) $X+Y \mid X = U \mid V$

$$\therefore f_{U|V}(u,v) = \frac{f_{U,V}(u,v)}{f_V(v)}$$

note $X = V \rightarrow bin(n,p)$

$$= \frac{\frac{n!}{v!(u-v)!(n-u)!} p^v q^{u-v} (1-p-q)^{n-u}}{\binom{n}{v} p^v (1-p)^{n-v}}$$

$$= \frac{\frac{\cancel{n!}}{\cancel{v!}(u-v)!(n-u)!} \cancel{p^v} q^{u-v} (1-p-q)^{n-u}}{\frac{\cancel{n!}}{\cancel{v!}(n-v)!} \cancel{p^v} (1-p)^{n-v}}$$

$$= \frac{(n-v)!}{(u-v)!(n-u)!} \frac{q^{u-v}}{(1-p)^{u-v}} \cdot \frac{(1-p-q)^{n-u}}{(1-p)^{n-u}}$$

$$= \binom{n-v}{u-v} \left(\frac{q}{1-p}\right)^{u-v} \left(1 - \frac{q}{1-p}\right)^{n-u}.$$

4. $\int_0^\infty \frac{\lambda^k x^{k-1}}{(k-1)!} e^{-\lambda x} dx$

$\lambda x = y \qquad x = \frac{y}{\lambda}, \quad dx = \frac{dy}{\lambda}$

$$= \frac{\lambda^k}{(k-1)!} \int_0^\infty \left(\frac{y}{\lambda}\right)^{k-1} e^{-y} \frac{dy}{\lambda}$$

$$= \frac{1}{(k-1)!} \underbrace{\int_0^\infty y^{k-1} e^{-y} dy}_{= \Gamma(k) = (k-1)! \ (\because k \in \mathbb{N})} = 1$$

or set $\alpha = k$ & $\beta = \frac{1}{\lambda}$,

then this is a gamma dist.

5. $f_X(x) = 1 \quad (0 < x < 1)$

$f_Y(y) = 10 \quad (0 < y < 1)$

$$Cov(X, Z) = E(XZ) - (EX)(EZ)$$

$$= E(X(X-Y)) - (EX)(E(X-Y))$$

$$= EX^2 - E(XY) - (EX)^2 + (EX)(EY)$$

$$= EX^2 - (EX)^2 - \underbrace{(E(XY) - (EX)(EY))}$$

$$= Var(X) = \sigma_X^2 \qquad = 0 \quad \because \text{indep.}$$

$$Var(Z) = Var X + Var(Y)$$

$$= \sigma_X^2 + \frac{1}{100}\sigma_X^2 = \frac{101}{100}\sigma_X^2$$

$$\therefore \rho_{X,Z} = \frac{Cov(X,Z)}{\sigma_X \sigma_Z} = \frac{\sigma_X^2}{\sigma_X \sqrt{\frac{101}{100}}\sigma_X}$$

$$= \sqrt{\frac{100}{101}}$$